

Demystifications and Reenchantments of the Humanlike Machine

Since the outset of the discipline, anthropology has been signally preoccupied with a series of problems to do with ostensibly peculiar relations between persons and ‘things’ which somehow ‘appear as’, or do duty as, persons.

(Gell 1998: 9)

As this epigraph from Alfred Gell suggests, the distinction of humans and nonhumans marks a relation that has been extensively explored not only by anthropologists but also by their research subjects, among other ways through the fungibility of persons and things. My interest in this book is to contribute to explorations of these “peculiar relations” at a particular site of contemporary cultural imaginaries, that is, the computing sciences and arts. I have posited that the fascinations of artificial personhood for AI practitioners involve a kind of mimesis that works as a powerful disclosing agent for associated assumptions about the human.¹ In this chapter I explore those assumptions through some specific encounters with contemporary humanlike machines, read through the lens of recent developments in the anthropology of science and technology and related fields.² Capacities for action are recast in these writings from

¹ Sherry Turkle (1984) was among the first to take up the broader question of the “aliveness” of computational artifacts and the workings of the computer as what she names an “evocative object” for the human. I use the term *mimesis* here in the sense developed by Michael Taussig, as “the faculty to copy, imitate, make models, explore difference, yield into and become Other” (1993: xiii).

² More specifically, my aim in this chapter and the next is to identify the outlines of a figure of humanness more consistent with conceptualizations currently under construction within the fields of cultural anthropology (Downey and Dumit 1997; Gell 1998; Strathern

inherent capabilities to possibilities generated and reiterated through specific sociomaterial assemblages and enactments. These approaches shift the frame of reference from the autonomous human individual to arrangements that produce effective forms of agency within ramifying networks of social and material relations. Just how those networks are drawn and “cut” (Strathern 1996), and with what agential effects, is a practical, political, and aesthetic question, as well as a materially consequential one.

For scholars in the humanities and the social sciences, projects of artificial life in its various forms conjoin with ongoing reexamination of the tradition of liberal humanism and its figurings of persons, agency, and the like. This project of articulating the “posthuman” has been extensively described by Hayles (1999, 2002, 2005), who sees initiatives in the fields of situated robotics and artificial life as indicators of more profound shifts in the human sciences and in contemporary society. The position that I develop below intersects with that of Hayles but also differs in what I believe are critical (in both senses of that term) ways. I embrace Hayles’s analysis of the history of the information sciences and the discursive disembodiment of mind and share her interest in the possibilities that computing affords for rethinking traditional conceptions of the human. But whereas Hayles anticipates resistance to projects in situated robotics and artificial life from those wedded to a conservative humanism (2005: 143), my own resistance to those projects is based on quite different grounds. I discuss these grounds in detail below, but in brief my concern is that, like their predecessors, these projects continue to restage the parochial and conservative forms of liberal humanism that in Hayles’s reading they leave behind.

At the same time, my own analyses of initiatives in the design of humanlike machines have been concerned with resisting too easy elisions of difference at the human–computer interface. My approach has been to slow down discourses of the “smart” machine to attend closely to the practices through which purportedly intelligent and interactive artifacts are realized, including just what conceptions of intelligence and interaction are in play. The result of this is an enduring skepticism regarding the rhetorics of machine intelligence and an interest in

1999; Taussig 1993), science and technology studies (Casper 1994; Collins 1990; Cussins 1998; Haraway 1991, 1997; Latour 1993, 1999; Law and Mol 2002; Lenoir 2002), feminist theory (Barad 2003; Braidotti 1994, 2002; Butler 1993; Halberstam and Livingston 1995), and cultural studies (Balsamo 2000; Doyle 2003; Featherstone and Burrows 1995; Hayles 1999, 2002, 2005; Kember 2003).

demystifying the specific technologies and practices about which these discourses make their claims.

With the critique of the [previous chapter](#) in mind, I turn here to particular encounters with several of the most highly acclaimed realizations of the intelligent machine, the situated robots Cog and Kismet. In offering some different readings of these projects from those available in popular media representations, I hope to indicate how encounters at the human–computer interface could support more radical reworkings of the figures of both. This alternative is further elaborated through the case of an artifact sited more provocatively at the intersection of AI and new media art, performance artist Stelarc’s *Prosthetic Head*.³ My interrogation of the *Head* expands the unit of analysis to incorporate the ongoing labors of design practitioners, the unruly contingencies and material particularities of computational artifacts, and the artfully collusive performances that make up encounters at the human–computer interface.⁴ In pursuing this line of research, I engage the body of scholarship that emphasizes the inseparability of the human from the artifactual and renders the relation as a more radically irreducible and intimate one. Taken together, these reflections are meant to contribute to the project that Taussig has named the demystification and reenchantment of lively things (1993: 1).

MYSTIFICATIONS AND ENCHANTMENTS

A close consideration of humanlike machines suggests a radical inversion of the premises that inform AI and robotics projects, of even the most “situated” kind. Rather than flawed approximations of autonomous agency, we can take these artifacts as demonstrations that

³ In her essay “Flesh and Metal: Reconfiguring the Mindbody in Virtual Environments” (2002), Hayles shifts her focus from AI and robotics to the computational arts, where she finds resources for a more radical rethinking of bodies, embodied experience, and the (post-)human. I share this assessment of where we might look for practices at the human–computer interface more aligned with recent critical reconceptualizations of persons and things.

⁴ Hayles’s discussion of the installation *Traces* is highly resonant with my argument here, in pointing to the designers’ deliberate avoidance of effects that would obscure the particularities of the computational medium and their orientation to the interactions between persons and the VR environment. On her account, the designers turned problems that they encountered in getting a computationally generated avatar to exactly track the user’s body into a lesson: “rather than regarding the avatar as a mirroring puppet, they think of it as a trace emerging from the borderlands created by the energetic body in motion. What was a tracking problem is thus transformed into the possibility of creative play between user and avatar” (2002: 308–9).

agencies – whether robotic or human – might need to be conceptualized on very different grounds. It is at this point that recent critical writings on subjects, objects, and relations between them become relevant for rethinking interactions between humans and machines. In *Strange Encounters*, for example, Sara Ahmed develops a critique of the figure of “the stranger” by extending Marx’s analysis of commodity fetishism (as the substitution of an enigmatic object for the social relations of labor) to include fantasy as well as materiality or a “fetishism of figures” (2000: 4). Among other moves, Ahmed argues, this form of fetishism involves cutting off a figure from the histories of its determination through particular, embodied encounters. The figure is treated as “having a nature,” as being something that *is* and that has effects, rather than as an effect in itself. Ahmed’s interest is not only in how such figures are produced but also in how they are put to work in particular times and places, including the labor that the fetishized figure conceals.

In an argument from a very different domain, but with strong resonances to Ahmed’s, Gell (1998) has explored the “enchantment” of objects brought about through the masking of labors of production. Like Taussig, Gell starts from Walter Benjamin’s proposal that mimetic practices, resulting in a plethora of images and simulacra, are based in a compulsion to imitate the world as a means of gaining access to it (see Gell 1998: 100; Taussig 1993: 20). Rejecting the idea of mimesis as a “primitive faculty” inherited from the past, Gell and Taussig adopt the category of mimesis as the more particular production of figures whose salient property is their resemblance to an original. They then use the concept to analyze a range of practices of magic and sorcery and their associated artifacts. Gell traces the history of this form of object enchantment back to Tylor, who, in *Primitive Culture* (1875), defined primitivism in terms of animism or the attribution of life and sensibility to inanimate things, as well as to Frazer (1900), Malinowski (1935), and Mauss (1902, 1954) in their studies of magic and the efficacy of objects in relations of exchange. Mauss’s theories of exchange, in particular, consider how gifts act as extensions of persons and inspire Gell’s theory of the agencies of art and of artifacts more generally. Artifacts are, by definition, Gell proposes, those objects taken to be instruments or outcomes of social agency. A found object like a stone, placed on a mantelpiece, becomes an art object and an artifact, indexing the agency of its finding and placement. At the same time, the enchanted object’s effects are crucially tied to the indecipherability of prior social action in the resulting artifact.

It is through the specific materialities of the artifact, crucially, that its effects, and the absent presences on which they depend, are achieved.

Taussig (1993) takes as his focus what he characterizes as “the two-layered notion of mimesis that is involved [in Benjamin’s analysis] – a copying or imitation, and a palpable, sensuous connection between the very body of the perceiver and the perceived” (1993: 21). From this he develops a critique of Frazer’s typology of sympathetic magic into the two categories of “imitation” and “contact,” arguing that the two are always intertwined in practice. These observations call out the importance of the particular materialities of objects to their efficacy, however much the latter is also based in ephemeral and intangible imaginings. Indeed, by this analysis the two aspects – embodied, sensuous contact and magical efficacy – are inextricably intertwined.

Developed with respect to fetish figures made of wood, this analysis is richly suggestive for thinking about the computationally powered artifacts of contemporary AI and robotics. A more general characteristic of object fetishism in an age of commodity capitalism, the entanglement of sensuous corporeality and apprehension of the liveliness concealed within things has particular resonance in the case of humanlike machines. This despite the fact that the humanness assumed in discussions of the potential success (or the inevitable failure) of attempts to replicate the human mechanically is typically stripped of its contingency, locatedness, historicity, and specific embodiments. If, in contrast, we take the human to be inseparable from our specifically situated social and material circumstances, the question shifts from “Will we be replicated?” to something more like “In what sociomaterial arrangements are we differentially implicated, and with what political and economic consequences?” This alerts us, in turn, to the possibility of encounters at the interface conceived very differently than as the meeting of a human and a machine, each figured as a self-standing entity possessed of preestablished capabilities. Rather, effective encounters at the computer interface are those moments of moving complicity between persons and things achieved through particular, dynamic materialities and extended socialities.

DEMYSTIFICATIONS

To make the preceding argument more concrete, I return to the particular humanlike machines introduced above. I was able to experience my own encounters with the robots Cog and Kismet on a visit to the MIT AI lab in the fall of 2001. Cog, as the graduate student who led our tour explained apologetically, was inactive and had been so for some time. As there were no researchers actively working on its development, Cog was afflicted

by a condition commonly known as “bit rot,” or the degradation of code in the absence of ongoing maintenance of its compatibility with continually changing software and hardware environments (see Smith 1996: 203). We were, however, able to visit the inanimate Cog sitting in a corner of the lab. Although still an imposing figure of a robot, what struck me most powerfully about Cog was the remainder of its “body” not visible in media portrayals. The base of Cog’s torso was a heavy cabinet from which came an extraordinarily thick sheaf of connecting cables, running centaurlike to a ceiling-high bank of processors that provided the computational power required to bring Cog to life. Seeing the robot “at home” in the lab, situated in this “backstage” environment, provided an opportunity to see as well the extended network of human labors and affiliated technologies that afford Cog its agency, rendered invisible in its typical media staging as Rod Brooks’s singular creation and as an autonomous entity.⁵

Although Kismet was operational, in contrast to the interlocutors pictured in the Web site videos, none of our party was successful in eliciting coherent or intelligible behaviors from it. Framed as an autonomously affective entity, Kismet, like Cog, must be said to have failed in its encounters with my colleagues and me. But as in the case of Cog, there are more interesting and suggestive lessons to be learned from the difference between Kismet’s demonstrated competencies and the Kismet that we encountered. Those lessons require that we reframe Kismet, like Cog, from an unreliable autonomous robot, to a collaborative achievement made possible through very particular, reiteratively developed and refined performances. The contrast between my own encounter with Kismet and that recorded on the demonstration videos makes clear the ways in which Kismet’s affect is an effect not simply of the device itself but of Breazeal’s trained reading of Kismet’s actions and her extended history of labors with the machine. In the absence of Breazeal, correspondingly, Kismet’s apparent randomness attests to the robot’s reliance on the performative capabilities of its very particular “human caregiver.”⁶ Like all forms of agency, in other words, Cog and Kismet’s capacities for action are created out of sociomaterial arrangements that

⁵ For a closely related observation in a very different case, see Miallet’s account of the extended corporeality of physicist Stephen Hawking (Miallet 2003).

⁶ See <http://www.ai.mit.edu/projects/sociable/overview.html> (last accessed November 7, 2005). In an interview (Menzel and D’Aluisio 2000: 71), Breazeal responds to the suggestion that Kismet might be tuned specifically to her as an interactional partner with the hope that this is an early stage of the robot’s development that will improve with time. One could argue, however, that this is the case with all beings both human and

instantiate histories of labor and more and less reliable, always contingent, future reenactments.

If humanlike robots comprise an other that mimes dominant cultural imaginaries of the human, a more literal doubling is materialized in performance artist Stelarc's *Prosthetic Head*.⁷ The head is a three-dimensional graphical simulacrum of Stelarc's own, endowed with the capacity to take queries typed on a keyboard and to respond in automatically generated speech.⁸ On exhibit at the InterAccess Gallery in Toronto in spring of 2003, the *Head* was displayed in larger than life-size dimensions on the wall of a darkened room, with no accompanying artifacts other than a pedestal holding a standard computer keyboard.⁹ My visit to the gallery on March 29, 2003, was motivated by an interest in comparing the *Head* with previous attempts at conversational agents. I was delighted to discover that on that day the original Stelarc was also on hand, observing encounters with his digital doppelganger.¹⁰ Our encounter, lasting over an hour and videotaped independently by my companion and by Stelarc himself, was augmented by Stelarc's

nonhuman, from the proverbial "two-year-old" child to the artifacts discussed throughout this book. In both cases, intelligibility is tied to increasingly intimate familiarity.

⁷ See <http://www.stelarc.va.com.au/prosthetichead/>. The artist credits the following collaborators for the creation of the *Head*: Karen Marcelo, project coordination, system configuration, alicebot customization (<http://karenmarcelo.org/>); Sam Trychin, customization of 3D animation and text to speech software; Barrett Fox, 3D modelling and animation (<http://www.barrettfox.com/>); John Waters, system configuration and technical advice (<http://www.shitech.net/>); and Richard Wallace, creator of alicebot and AIML (<http://alicebot.org/>) (Stelarc 2004).

⁸ It is an interesting feature of conversational agents like the *Head* that they make evident the greater contingency of competent machinic hearing than of speaking. Although automatic speech generation is a relatively mature and robust technology, speech recognition still requires prolonged "training" of machines to the particularities of individual voices. This affords an interesting contrast to human language learning, where comprehension is widely experienced to precede fluent expression.

⁹ Stelarc reports that the database that informs the *Head* has since been purchased by the Australian National Gallery in Canberra, The Sherman Galleries in Sydney, and Gilbert Silvermann, an American collector in Detroit. After its exhibition at the InterAccess Gallery, the *Head* was installed as a four-meter-high display and exhibited at ACMI (The Australian Centre for the Moving Image) in Melbourne. (Stelarc, e-mail communication, August 22, 2005.)

¹⁰ Taussig (1993), in a chapter titled "The Talking Machine," remarks on the fascination of European colonists with the fascination of "primitive" peoples with the former's technologies, "the white man's fascination with the Other's fascination with white man's magic" (207), including most dramatically the phonograph. He cites the number of accounts of "first contact" between native peoples and technologies, as retold by observing Europeans. On this analysis artifacts like the *Prosthetic Head*, and I would argue unfamiliar technologies more generally, are aimed at rendering us all as primitives as the creators of these machines look on with fascination.



FIGURE 14.1. The author in conversation with Stelarc's Prosthetic Head, Inter Access Gallery, Toronto, Canada 29 March 2003 (photograph courtesy of Andrew Clement).

responses to the *Head's* responses to my queries, as well as by his ongoing commentary on the *Head* and advice on how best to engage with it.

An encounter with Stelarc's *Head* affords further evidence for the collective and contingent nature of sociomaterial agencies. I offer one indicative exchange:

Stelarc: It also has a simple way of generating poetry on the fly, it has a simple database with keywords, you can say, "recite a poem" (pause)

LS: (types) "Recite a poem, please"

Stelarc: It'll, basically put together a bunch of words tha-

Head: [Your polite style is very nice.

Our breathing imploding breathing imploding,
City body electric system city excessive replicating,
And city city involuntary imploding. (Smiles)

LS: Oh, that was very nice! (laughs, glance to Stelarc, types) "Very nice!"

- Head: Thanks, still Lucy.
LS: (types) "Recite another one."
Head: Bodies not the involuntary so exchanging,
Absent inconsequential excessive perhaps city blurring,
Pulsing system electric excessive obsessively. (Smiles)
LS: (laughs) Yeah, right on.
Stelarc: (laughs) And it does have some interesting kind of philosophical kind of data, and information,
LS Yeah?
Stelarc: and definitions, umm . . .
Obs: Can it recite any philosophers, or commentators?
Stelarc: Well for example if you ask it "What is reductionism" (laughter)
Stelarc: That's one of it's favorite, my favorite, ah . . .
LS: (types) "What is reductionism?"
Head: Reductionism is the philosophy that all psychology reduces to biology, all biology to chemistry, chemistry to physics, physics to mathematical logic. Therefore, according to reductionism, I can understand you by logic alone without having a human brain.
Obs: (laughs) What about emergent properties?
LS: (types) "Do you believe in reductionism?"
Head: Yes I believe in it. Do you?
LS: (laughs, types) "I'm not sure."
Head: Is there anything I can say to help you make up your mind? (Much laughter all around, including Stelarc)
LS: (types) "Do yo" (backspaces to begin over)
Obs: Why doesn't he have feelings?
LS: (turns around to Obs) What?
Obs: Why don't you have any feelings?
LS: (types) "Do you have feelings?"
Head: As a robot I do not have any human emotion. My apparent emotional responses are simply algorithms designed to make me easier to interact with.
LS: (types) "Do you think robots should have emotions?"
Head: That depends on what you mean by thinking.

This sequence is readable for some recurring characteristics of interactions with conversational software. In particular, near the beginning is a moment that reveals a basic, and clearly consequential, difference

among the human and nonhuman participants in the encounter. I am referring to the *Head's* “interruption,” in its first turn, of Stelarc’s account of its poetic abilities, indicative of its lack of any awareness of other events in the room around it. As I have elaborated in previous chapters, human–computer interactions occur at a relatively narrow point of intersection, that is, at just those moments when the human participant takes an action that changes the system’s state (in this case, by hitting the return key on the keyboard). In contrast, human interactions invariably exceed what is made accountable in any direct sense. And it is this contingent horizon of possibly relevant phenomena, I want to propose, that effects interaction’s open-endedness.¹¹

I return to my exchange with Stelarc’s *Head* shortly but want first to pursue for a moment the question of open-endedness and the cybernetic. In “Cybernetics and the Mangle” (2002), Andrew Pickering explores the resonance that he sees between the work of mid-twentieth-century British cyberneticists Hal Ashby, Stafford Beer, and Gordon Pask and Pickering’s own theorizing of practice as “the emergent interplay of human and material agency” (ibid.: 414; see also Pickering 1995). He offers as one realization Ashby’s homeostat, a device designed to achieve self-regulation through an iterative succession of autoreconfigurations – a process effecting what Pickering characterizes as a form of “liveliness.” He writes: “I can’t actually think of any prior example of a real machine that would randomly – *open-endedly*, as I would say – reconfigure itself in response to its inputs . . . It seems reasonable, then, to speak of the homeostat as having a kind of *agency* – it did things in the world that sprang, as it were, from inside itself, rather than having to be fully specified from outside in advance” (ibid.: 417, original emphasis).

I want to focus on the elision in this comment on Ashby’s homeostat between randomness and open-endedness and also on the association

¹¹ In an article developed as a contribution to discussion of “awareness” within the field of CSCW, Heath et al. present a series of studies that make evident “the ways in which participants design activities to have others unobtrusively notice and discover actions and events which might otherwise pass unnoticed” (2002: 317). They make the important point that the mutuality of awareness does not necessarily involve a sameness or even symmetry among participants’ orientations to an ongoing course of action: it is an orientation to the contingent possibility that something might become relevant that is crucial. It is the capacity for the latter, rather than any assumed sameness, that my use of “asymmetry” in the case of the human–machine interface is meant to reference. Heath et al. point out that “awareness” does not mean some form of general orientation through which relevant events are filtered but rather very particularly selective attention to an environment that, in turn, is highly differentiated. See also Pedersen and Sokoler (1997), Robertson (2002).

between agency and something that comes from “inside.” First, what about the relation between randomness and open-endedness? Pickering here treats them as synonymous, though he concedes further on that in the case of the homeostat it is an open-endedness of a delimited kind, however large the space of possibilities. His idea of the mangle, in contrast, turns on a picture of agency as “indefinitely” open-ended.¹² The agency of the homeostat, moreover, was in an important sense a prespecified one; that is, “there was a principle of stability hard-wired in.” In contrast, Pickering observes: “The mangle is in at least two ways on the wild side of the homeostat – involving indefinitely open-ended searches of spaces of agency, and with no fixed principle of assemblage” (ibid.: 418). This leads him to call for an exploration of differences as well as similarities between instantiations of sociomaterial agency, a point that I return to below. But Pickering concludes with a celebration of cybernetics as a radical alternative to the classical sciences: “While the latter seek to pin the world down in timeless representations, cybernetics directly thematizes the unpredictable liveliness of the world, and processes of open-ended becoming” (ibid.: 430).

My own much more skeptical reading of the cybernetic project hangs on contradictions between the legacy of behaviorism, teleology, and control engineering that I believe still sits at the core of its devices and aspirations and a sense of contingency and interactivity worthy of the name of “unpredictable liveliness” and “open-ended becoming.”¹³ To pursue these questions, I turn back to my encounter with Stelarc’s *Head*. I was speaking of the *Head*’s (apparent) interruption of its progenitor, which in a moment of ironic serendipity takes the form of the *Head*’s complimentary remark regarding my own politeness. Rather than being read as rudeness, the *Head*’s action demonstrates its inability to perform the kind of competent interruption that rudeness requires, displacing a demonstration of social awareness with a mark of its machinic nature.¹⁴ A second moment of mechanism revealed occurs when the *Head* responds to my compliment on its poetic abilities by addressing

¹² Pickering is excited by the possibility that a particular class of cybernetic machines (for example, Ashby’s “Musicolor” device) can serve as instantiations of what he names “decentred becomings” (personal communication, November 8, 2005). The latter idea, as developed in his trope of “the mangle,” is, I believe, deeply resonant with the forms of reconfiguration that I consider here and in Chapter 15. See Pickering (1995, 2002).

¹³ On the history of cybernetics, including its universalizing and “closed world” rhetorics, see Bowker (1993), Edwards (1996).

¹⁴ On the extraordinary competences involved in conversational turn-taking, see, for example, Sacks et al. (1978), Goodwin (1981).

me as “still Lucy.” The installation at the InterAccess Gallery was configured with a pressure sensitive pad placed at the foot of the pedestal on which the keyboard rested: it is the visitor’s step onto the pad that activates the software running the *Head*, effectively “waking it up” and initiating a request for the visitor’s name.¹⁵ A standard feature of so-called conversational agents, this salutation effects what Chesher (2004) names an “avocation,” a kind of interpolation of the prospective user-participant into the configuration of the interface. In the case of the *Head*, the three-dimensional corporeality of the encounter is enhanced by the fact that, rather than sitting at a standard display screen, the visitor stands opposite the *Head*, which in turn fills her field of view. The activation of the *Head* software by the pressure-sensitive pad sets up a spatial conjoining of *Head* and visitor in which both are entrained, in other words, and which marks as well the temporal bounds of a particular exchange. On the day of our encounter, however, the pressure pad was repeatedly failing to transmit, with the result that the *Head* was liable to shut down and no longer respond to keyed input, in a way that would be appropriate had its interlocutor walked away. The fix for this bug was for the visitor to step off of the pad and then step back onto it again. This interrupted the continuity, however, with the result that the *Head* treated the ensuing exchange as a new encounter, once again requesting a name. To draw attention to this failure of recognition I introduced myself on reentering the dialogue with the phrase “I’m still Lucy” and was named “still Lucy” from that time on.¹⁶

Whereas breaches like this reveal the machinic limits of the *Head*’s interactional competencies, other moments effect an uncanny sense of presence and of generative spontaneity. The *Head*’s poetic abilities, as Stelarc explains, are based on serendipitous juxtapositions from an

¹⁵ The eyes of the *Head* when not activated are closed: the pressure on the pad initiates its processes in much the same way that a touch of the space bar awakens a sleeping laptop.

¹⁶ Stelarc reports that in its subsequent installation at the The Australian Centre for the Moving Image in Melbourne approximately 1,500 people went through the exhibition every day for five months, and the *Head* performed flawlessly (Stelarc, e-mail communication, August 22, 2005). In any case, I cite this breakdown not as anything extraordinary or as diminishing of the *Head*’s success. On the contrary, I would argue that in important respects it is moments like this which reveal the extraordinary achievements of technical systems. In breaching the general invisibility of the infrastructure that would otherwise go unnoticed, they call our attention to other actors – human and nonhuman – outside of the frame. I return to this question of framing below.



FIGURE 14.2. Prosthetic Head, San Francisco, Melbourne 2003, Programmers Karen Marcelo, Sam Trychin, 3-D models Barrett Fox, reprinted with permission from STELARC.

evocatively seeded database of “keywords.” What is not available from the transcript is the singularly enchanting performance of these verses, effected through the cadence of the *Head*’s speech and the dynamic animation of its face, particularly the eyes and mouth. Special attention has been paid in the design of the *Head* to the eyeballs, teeth, and tongue, each of which are separate moving elements in the 3000-polygon mesh model of the artist’s head from which the *Head* is constructed.¹⁷ This brings us to the second issue raised by Pickering, regarding agency as action initiated “from inside.” Pickering here invokes a contrast between things that achieve agency from within themselves or through external specifications. In his consideration of objects endowed with human (or superhuman) properties, Gell points out that mind can only ever be depicted suggestively, as that which is hidden “inside” an observable body. In the case of traditional idols, this is typically done through the introduction of some kind of opening or orifice, which designates an exterior–interior distinction to which the opening gives access. Eyes are the canonical example, as “the windows to the soul” (1998: 132, 136). In this sense mind and spirit are anthropologically symmetrical alternates, each standing for an indexically constituted mind–body contrast, an interior indicated by the surfaces of the body that enclose it. Gell observes: “There are thus two basic strategies for converting (conceptually) stocks and stones into quasi-persons in artifact-form. The first of these strategies consists of animating the idol by simply stipulating for it a role as a social other. The second consists of providing it with a homunculus, or space for a homunculus, or turning it into a homunculus within some larger entity” (1998: 133). Gell points to the primacy of vision and of the eyes in Hindu iconography, union coming from eye contact with the gods, as the “eyes of the god, which gaze at the devotee,

¹⁷ See <http://www.acmi.net.au/7E8A5C8E6F304A839116C3C74F81440C.htm>.

mirror the action of the devotee, who gazes at the god” (1998: 119). Stelarc himself sees his project as a direct challenge to conceptions of consciousness, mind, intelligence, and the like as forms of mysterious interiority. In the context of prevailing figures of humanness, however, the desire to effect an uncanny liveliness for the *Head* inescapably suggests those very things.

The poetry that the head recites involves an ingenious use of the stored database of keywords strung together in ways that exploit the often deliberately nonlinear juxtapositions of poetry and the irrepressibly suggestive and meaningful nature of language. One result of this is the generation of what Stelarc characterizes as “alternate, intimate, involuntary experiences” for the *Head*’s interlocutors.¹⁸ The flowing cadence of the *Head*’s recitation contributes to the sense of its artistry, with the result that its poetry generates laughter and delight from its listeners. Similarly, the exchange regarding the nature of “reductionism” reverberates with its reflexive relevance for the *Head* as speaker and comprises a moment of astonishingly coherent alignment effected by the well-established technique of matching certain key phrases (“I’m not sure”) with associated responses (“Is there anything I say to help you make up your mind?”).¹⁹ What affords the liveliness of the exchange

¹⁸ See <http://www.stelarc.va.com.au/index2.html>.

¹⁹ The *Head* is based on the long-standing and widely circulating platform called ALICE. See <http://www.alicebot.org/>. (See Chapter 12.) Stelarc explains that the *Head*’s “general knowledge” is based on the ALICE database, which has had over five hundred contributors since 1995 when “Alice the Chatterbot” and AIML (the Artificial Intelligence Mark-up Language) was initiated by programmer Richard Wallace. The *Head*’s database has “inherited” much of this from ALICE but has also, since late 2002, been altered and extended, as Stelarc explains:

in order to personalize the Head, to more express the artist’s concerns and concepts. Not only in what it knows and how it says it but also that it develops some of its own “creative” capabilities. For example, the Head can make poetry-like sentences and song-like combinations of sounds (generated differently each time it is asked). Much of the Alice data-base is neutral information about the world, but some of it expresses the peculiar beliefs of the programmers. Often exposing particular American biases (the right to bear arms) and certain religious beliefs (a Christian God). Although much of this kind of data has been deleted or adjusted, occasionally the Head gives conflicting answers. It doesn’t believe in God (me!) but it might still have embedded somewhere a response that contradicts this. So it is a fundamentally schizoid entity. (Aren’t we all?) (Stelarc, e-mail communication, August 25, 2005)

This complex and inconsistent “inheritance,” as Stelarc characterizes it, is part of what makes up the *Head* just as ours makes us.

is the simultaneous recognition of that alignment, of the creative programming labors that stand behind it, and of the ways in which it points, unwittingly, to a future relationship – of persuasive discussion – that we anticipate actually exceeds the *Head's* capacity.²⁰ That mix of alignment and slippage, and the response that it evokes, is demonstrated most dramatically in the close of the sequence. The question “Do you think that robots should have emotions?” again explores the bounds of the *Head's* self-reflective and humanlike capacities. But where the question anticipates an orientation to the problem of emotion, the *Head* disarmingly organizes its response around a prior problem – one moreover that is defining of the project of which the *Head* is a part – that of thought itself.

Reflecting on my afternoon with Stelarc and his *Head*, I am struck by the sense of collaborative performance involved, both within and beyond the gallery walls. Within the encounter all worked together to make the particular assemblage of exhibition space, persons, and artifacts cohere into something if not always intelligible, at least interesting and worthwhile. As Kember characterizes the progeny of researchers in artificial life, “these novel agents do not so much evolve as coevolve in the dynamic interplay between observer and object, and they are more a facet of communication – the desire for life – than of computation – alife itself” (2003: 6). Stelarc is interested in what he describes as the “seductive couplings” that occur between the head and its interlocutors. He playfully proposes as well that the *Head* might serve a more useful purpose: of standing in for the artist in response to requests from the media, eager students, and the like. Framed as his virtual double, Stelarc's *Prosthetic Head* does effectively distribute the artist's agency. It does so not through any simple form of replication, however, but as an effect of complex and shifting assemblages of persons and things, specifically situated in time and place. It is to the question of differences *within* such assemblages that I now turn.

²⁰ Stelarc explains that because of the limitations of the AIML programming language, the *Head* is unable to increase its database from the conversations that it has. “It is programmed in stimulus–response modules. You anticipate the queries, you provide data for its responses” (Stelarc 2004). On Stelarc's view it is this inability to learn from its conversations that prevents the *Head* from being an artificial intelligence. At the same time “as its database increases the head will become more informed and less predictable in its responses. The head will appear to be more autonomous. The artist would then no longer be able to take full responsibility for what his head says” (Stelarc 2004).

REENCHANTMENTS

These reflections on encounters at the human–computer interface are meant to contribute to an awakening of the “congealed life in the petrified objects” of AI and robotics (Taussig 1993: 1). Contrary to the apparent enlivening of objects promised by the sciences of the artificial, I want to propose that an investment in obscuring the performative foundations of persons and things instead works to deaden the resulting artifacts. To further emphasize these foundations of subject and objecthood, I turn to the question posed by Marilyn Strathern regarding the personification of objects. Strathern approaches the answer by asking a prior question; that is, in what respects might we understand the constitution of persons as a form of reflexive objectification?²¹ She takes as her frame associated acts of separation and relation, as “it is through the separation of persons from one another that specific relations are created, and through relations that persons are defined . . .” (1999: 16). In her earlier studies of Melanesian personhood, Strathern introduces a distinction between “person” and “agent.” The “person” is not a preexisting entity but an object of the regard of others and an objectification of the relations that constitute her. The “agent,” in turn, is the one who acts with those relations as cause and reference – “with another in mind” (1988: 272–4).

The project of creating a humanlike machine requires that the phrase “with another in mind” be questioned more closely, however, than its colloquial sense would suggest. Standard readings within cognitive science would attempt a kind of literal mapping into a model of the mind of one actor of some model of the other. But the context of Strathern’s statement suggests something else. The mind of the Hageners as Strathern recounts it is not an entity contained within the person but rather the enactment of an elaborate, and elaborating, history of social relationships implying specific agencies and consequences. It is impossible, on Strathern’s account, to conceptualize Melanesian agency apart from those relations: agents are not the authors of their own acts, albeit that their actions are no less their own. This argument suggests that intentionality needs to be understood not as an attitude of mind located within the individual but as a field of socially and materially mediated relations within which persons act.

²¹ Working within a very different intellectual tradition, social philosopher George Herbert Mead makes a related argument in his classic exploration *Mind, Self and Society* (1934).

These arguments reverberate with others in cultural anthropology as well, located at the interface not between humans and machines but at the moments when humans come into, and go out of, recognized social personhood. Casper (1994, 1998), Franklin (2000), Hogle (1999), Lock (2002), Thompson (Cussins 1998; Thompson 2005), and others concerned with the anthropology of birth and death have drawn our attention to the ways in which these transition zones at the edges of life provide poignant sites for the study of personhood and its problematics. In a moving analysis of neonatal intensive care, Middleton and Brown (2005) cite a physician who observes that at some point in the process of working to keep a premature baby alive it is the baby who, as the physician phrases it, “decides” its future. Read through the lens of Strathern’s discussion of “cutting the network” (1996), this observation can be understood not as a statement of the infant’s acquisition of autonomous agency but as a practical move in the enactment of the sociotechnical networks of which the infant is an integral part. The account by Middleton and Brown describes the ever-ramifying extensions to the network of neonatal intensive care that are now possible: as long as more and more human and nonhuman resources can be brought to bear, the infant body can, at least potentially, be sustained. Rather than reading the physician’s statement to say that the infant at some point takes over the agency of this collective, we might read it as a statement that at some point the autonomy of the infant *must be posited* for the collective effort, and ramifying extensions to the network, to be cut. Constituting the infant’s autonomous agency, in other words, makes it possible at least partially to disengage this morally and emotionally central – at once fragile and powerful – entity from its reliance on the wider human–nonhuman collective.

The sciences of the artificial operate in a cultural and historical frame that takes autonomous agency not as an effect of cutting the network, however, but as the precondition for participation in it. Given an ontology of separate things that need to be joined together, machines must in some sense be granted agency to be brought into relationship with us. Given a model of humanness as a set of separate components that need to be incorporated into a coherent whole, and the resulting individual as an entity that needs to be acculturated into a set of social relations, the projects of artificial intelligence make some kind of sense. But what if our starting place comprises configurations of always already interrelated, reiterated sociomaterial practices? What if we understand persons as entities achieved only through the ongoing enactment of

separateness and always in relation with others? Rather than working to create autonomous objects that mimic Cartesian subjects, we might then undertake different kinds of design projects – projects discussed in the following chapter under the theme of human–machine reconfigurations.